OBJECT ORIENTED PROGRAMMING LAB

ASSIGNMENT-9

(Topics – Templates in C++)

Programming Questions

1. Write a function template maximum() that accepts two values and returns the greater value. Test it with integers and floating-point numbers.

2. Write a function template swapValues() to exchange two values. Demonstrate it using integers and characters.

3. Implement a class template Calculator that stores two values and provides functions to perform addition, subtraction, multiplication, and division.

4. Create a class template ArrayOperations that stores an array of elements and provides functions to display the elements and find the largest element.

5. Create a class template Pair that stores two values, which may have different data types. Implement functions to display both values. Test it with an integer and a string.

6. Implement a class template Stack using an array. Include functions push(), pop(), and display(). Handle stack overflow and underflow conditions.